

Lab 4: Normal-Normal Conjugacy, Posterior Predictive Distribution and Monte Carlo

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1. Normal-normal posterior for coffee-bag fill weights

A coffee-roasting company fills bags labeled as containing 500 grams. An inspector measures the fill weight of 12 bags:

```
y <- c(503, 498, 501, 505, 499, 502,  
       500, 504, 497, 501, 503, 499)  
  
n <- length(y)  
ybar <- mean(y)
```

Let θ be the unknown mean fill weight. Assume

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \mathcal{N}(\theta, \sigma^2), \quad \sigma = 4,$$

and

$$\theta \sim \mathcal{N}(m_0, \tau_0^2), \quad m_0 = 500, \quad \tau_0 = 6.$$

The normal-normal posterior is

$$\theta \mid y \sim \mathcal{N}(m_n, v_n),$$

where

$$v_n = \left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2} \right)^{-1}, \quad m_n = v_n \left(\frac{n\bar{y}}{\sigma^2} + \frac{m_0}{\tau_0^2} \right).$$

Exercise 1: Posterior calculations

Question 1

Calculate n and $\bar{y}$.

```
cat("n =", n)
```

$n = 12$

```
cat("ybar =", ybar)
```

$ybar = 501$

Question 2

Calculate the data precision, prior precision, posterior precision, m_n , and v_n .

Question 3

Write the complete posterior distribution of $\theta \mid y$.

Question 4

Calculate the posterior mean, posterior variance, posterior standard deviation, and central 95% credible interval.

Question 5

Calculate

$$\Pr(\theta > 502 \mid y).$$